**OPERATING SYSTEMS (UE24CS242B)**

**Unit 1 – Introduction & Computer System Organization**

**Question Bank**

## Lecture 1 – Introduction

1. How do operating system objectives change when shifting from batch systems to interactive systems?
2. Explain why modern operating systems must balance hardware utilization with user experience.
3. Describe how the OS acts as an intermediary between application programs and computer hardware.

## Lecture 2 – OS Structure and Operations

1. Explain why protection mechanisms are essential in a multi-user operating system.
2. How does time-sharing improve CPU utilization in a multiprogrammed environment?
3. Discuss the role of hardware timers in preventing CPU monopolization.

## Lecture 3 – Kernel Data Structures

1. Why is efficient data structure selection critical for kernel performance? Give reasons.
2. Explain how operating systems manage shared resources in distributed environments.
3. Compare general-purpose operating systems with real-time operating systems based on task deadlines.

## Lecture 4 – OS Services, Design and Implementation

1. Explain how operating systems provide controlled access to hardware resources.
2. Why is modular OS design preferred over monolithic design in modern systems?
3. Discuss the importance of system logs and monitoring services in operating systems.

## Lecture 6 – Process Concept

1. Differentiate between a program and a process with suitable justification.
2. Why must each process be uniquely identified by the operating system?
3. Explain how multitasking improves system responsiveness.

**Lecture 7 – Process Management & I/O Lifecycle**

1. Why does process creation involve duplication of process attributes?
2. Explain how the operating system ensures fairness among competing I/O requests.
3. What problems can arise if processes are not properly synchronized during I/O operations?

**Lecture 9 – CPU Scheduling**

1. Why is CPU scheduling necessary in systems with both I/O-bound and CPU-bound processes?
2. Explain how scheduling policies affect system responsiveness and throughput.
3. Discuss why scheduling decisions must be fast and efficient.

**Lecture 10 – CPU Scheduling Algorithms (Part 1)**

1. Explain why simple scheduling algorithms may fail in real-world operating systems.
2. What factors should be considered when designing a CPU scheduling algorithm?
3. Discuss how arrival time and burst time influence scheduling performance.

**Lecture 11 – CPU Scheduling Algorithms (Part 2)**

1. Why is fairness an important consideration in scheduling algorithms?
2. Explain how time-sharing systems prevent process starvation.
3. Discuss the trade-off between response time and overhead in preemptive scheduling.

**Lecture 13 – CPU Scheduling Algorithms (Part 3)**

1. Why are multiple ready queues used in advanced scheduling algorithms?
2. Explain how feedback mechanisms help adapt scheduling behavior dynamically.
3. Discuss challenges involved in scheduling processes on multi-core systems.

**Lecture 14 – CPU Scheduling Case Study**

1. Why do modern schedulers emphasize fairness rather than strict priority ordering?
2. Explain how scheduler design impacts desktop system interactivity.
3. Discuss the challenges of scheduling in systems supporting virtualization.

**Lecture 15 – Introduction to Shell and Cron**

1. Why is the shell considered a powerful user-level program rather than part of the kernel?
2. Explain how shell scripting helps automate system administration tasks.
3. Discuss the advantages of using cron jobs over manual execution of recurring tasks.